Advanced Machine Learning Assignment 3: Time-Series Data – Summary Report
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Introduction: This assignment focuses on applying Recurrent Neural Networks (RNNs) to a time-series forecasting problem using the Jena Climate dataset. The primary objective is to predict future temperature values based on historical weather data and evaluate how different deep learning models perform on this task.

Time-series forecasting is challenging because it requires capturing temporal dependencies and patterns over time. This assignment explores multiple architectures, including Dense, Convolutional, and Recurrent Neural Networks, and compares their performance using Mean Absolute Error (MAE).

Dataset Description: The Jena Climate dataset contains weather observations recorded over several years. Each record includes multiple atmospheric features such as:

- Temperature (°C)
- Atmospheric pressure (mbar)
- Humidity (%)
- Wind velocity (m/s)
- Other related climate variables

Data Preparation: The dataset consists of 420,000+ time steps
The data was split into:

- Training set: 50%
- Validation set: 25%
- Test set: 25%

Normalization: To ensure stable training:

- Mean and standard deviation were computed from the training set only
- All data was normalized using these values
- This avoids data leakage and improves model performance

Methodology:

Time-Series Problem Setup:

The dataset was transformed into a supervised learning problem using a sliding window approach:

- Sampling rate: 6 (1 sample per hour)

- Sequence length: 120 (5 days of past data)
- Prediction target: Temperature 24 hours in the future

Each input consists of:

- 120-time steps of past observations

Each output corresponds to:

- Future temperature after 24 hours

Baseline Model: A naive baseline model was implemented where:

Future temperature = Current temperature

- Validation MAE ≈ 2.44
- This provides a reference for evaluating deep learning models

Models Implemented:

Dense Model:

1. Flattened time-series input
2. Fully connected layers
3. Limitation: Does not capture temporal dependencies

1D Convolutional Model:

1. Conv1D + pooling layers
2. Captures local temporal patterns
3. Limitation: Cannot model long-term dependencies effectively

LSTM Model:

1. Use memory cells to retain past information
2. Captures long-term dependencies
3. Significant improvement over Dense and Conv1D

LSTM with Dropout:

- Added dropout for regularization
- Reduces overfitting
- Improves generalization

Stacked GRU Model:

1. Two GRU layers stacked
2. More efficient than LSTM
3. Captures complex temporal relationships
4. Achieved best performance

Bidirectional LSTM:

1. Processes data forward and backward
2. Limitation: Not ideal for forecasting (future data not available)

Conv1D + LSTM Hybrid Model:

1. Combines:
    - Conv1D → short-term patterns
    - LSTM → long-term dependencies
2. Moderate improvement but not best

Model Evaluation:

All models were evaluated using:

- **Loss Function:** Mean Squared Error (MSE)
- **Metric:** Mean Absolute Error (MAE)

Model selection was based on:

- **Lowest validation MAE**

Final evaluation was performed on the test dataset.

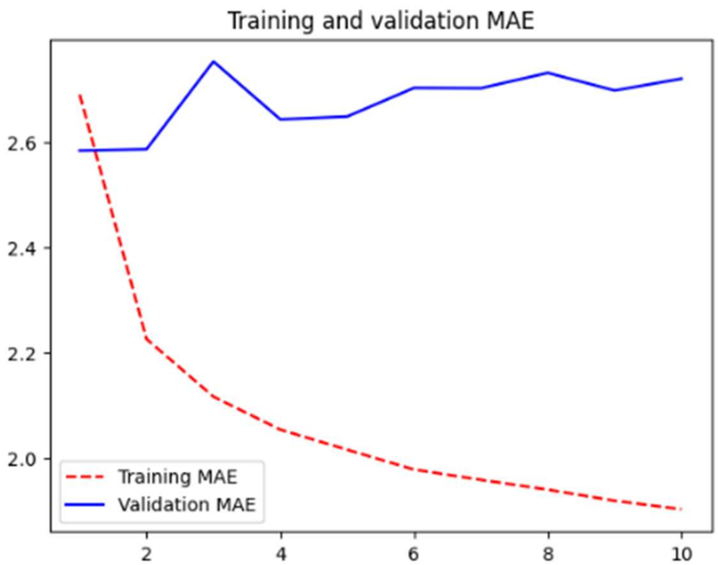
"The best-performing model was selected based on the lowest validation MAE".

Results Summary:

Model	Validation MAE
Baseline	2.44
Dense	2.58
Conv1D	2.91
LSTM	2.31
LSTM Dropout	2.31
Stacked GRU	**2.24**
Bidirectional LSTM	2.43
Conv1D +LSTM	2.50

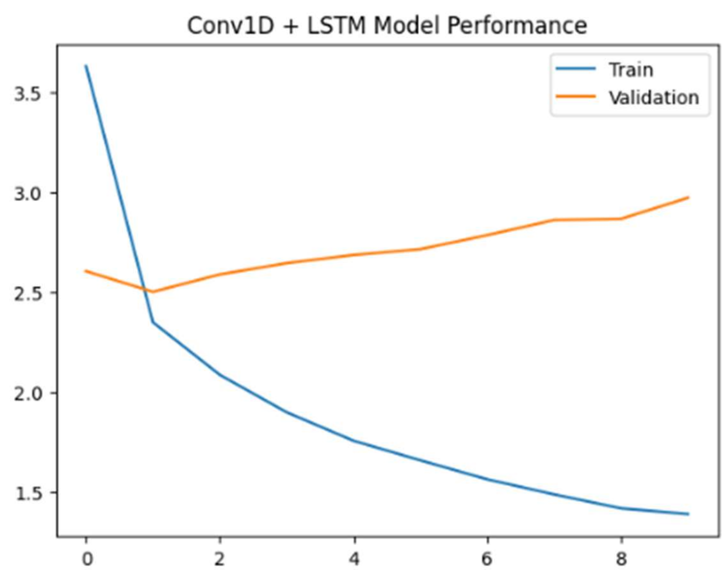
Graphs/Plots:

Training and Validation MAE:



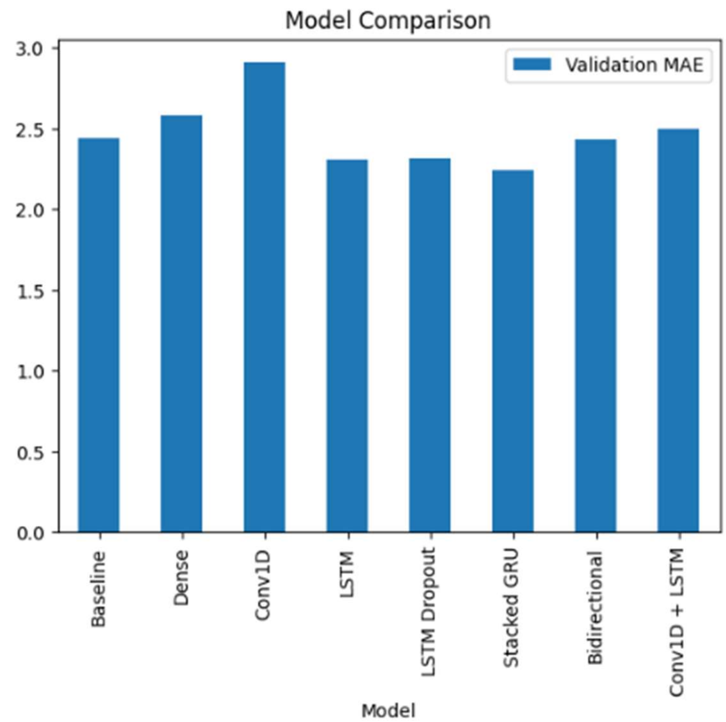
“This graph compares training and validation MAE over epochs, demonstrating the learning behavior and generalization of the model.”

Conv1D + LSTM Model Performance:



“This graph compares training and validation MAE over epochs, demonstrating the learning behavior and generalization of the model.”

Model Comparison:



"This graph compares validation MAE across all models. The Stacked GRU model achieved the lowest MAE, indicating the best performance".

Discussion:

- Dense models performed poorly because they ignore sequential relationships
- Conv1D models captured short-term patterns but failed on long-term dependencies
- LSTM significantly improved performance by modeling temporal dependencies
- GRU performed slightly better than LSTM due to simpler architecture and efficient learning
- Dropout helped reduce overfitting
- Bidirectional models are less suitable for forecasting tasks
- Hybrid models improved performance but did not outperform GRU
- The results indicate that increasing model complexity improves performance up to a certain point, after which gains become marginal.

Conclusion: The results demonstrate that recurrent neural networks are highly effective for time-series forecasting. Among all models, the stacked GRU achieved the lowest validation MAE, indicating the best performance in predicting future temperature. This experiment shows that simpler models fail to capture long-term dependencies, while GRU provides an efficient balance between accuracy and computational complexity.

Overall, models that incorporate temporal learning, such as LSTM and GRU, significantly outperform Dense and Conv1D models. This highlights the importance of sequential modeling in time-series data and confirms that RNN-based architectures are well-suited for capturing complex temporal patterns.